PAPER AIRCRAFT PROJECT

Design, Aerodynamics and Flight-Time Experiment

Objective

The aim of this project is to make a paper airplane using one A4 sheet and try to keep it in the air for as long as possible. The target is flight duration, not the longest distance. I made a broad-wing paper glider with a pointed front and a folded centre body.

Materials and rules

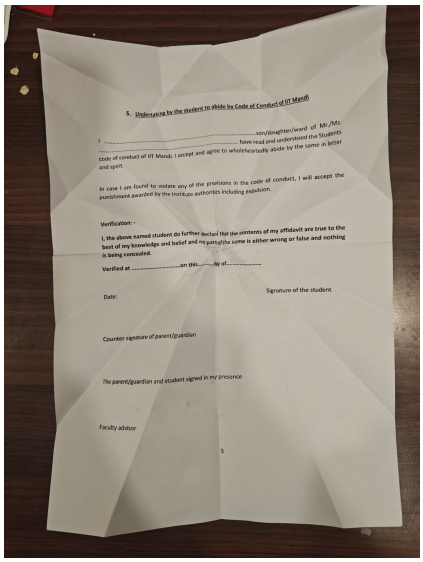
I used one standard A4 sheet of paper. The airplane was made by folding the sheet. No tape, glue, paper clips or motor were used. The plane is launched by hand and should be tested indoors or in a place with very little wind.

Part A — Design

1. How I made my airplane

Step 1 — First folds

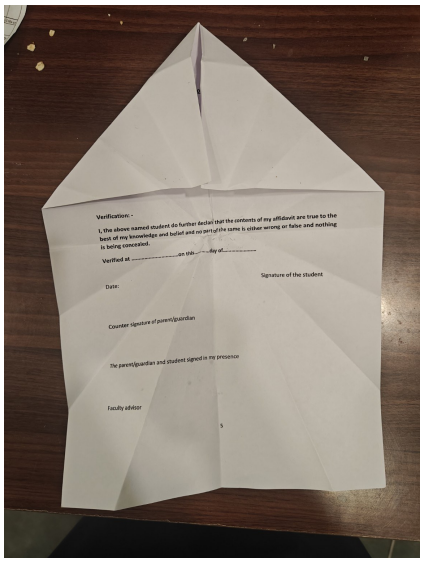
I started with the A4 sheet and made the first folds to create a centre line and guide the rest of the design.



My construction photo — Step 1

Step 2 — Pointed front

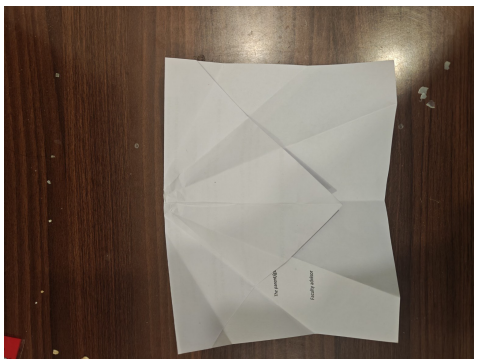
I folded the top corners inwards to make a pointed front section.



My construction photo — Step 2

Step 3 — Centre body

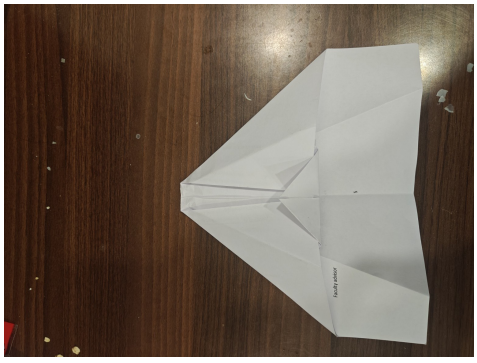
I made more folds near the middle to form a narrow central body and keep the front section together.



My construction photo — Step 3

Step 4 — Main wings

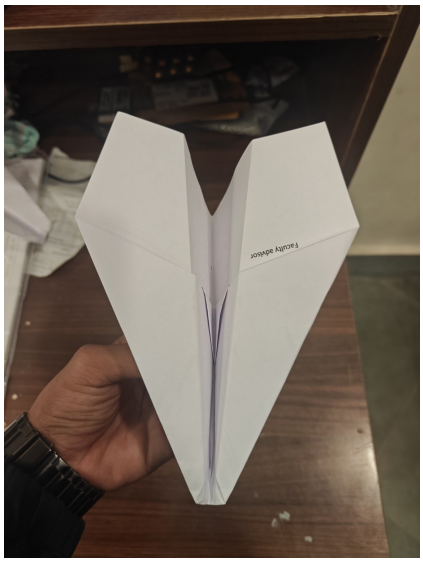
I folded the large side sections to make the wide, triangular wings.



My construction photo — Step 4

Step 5 — Final shape

I checked the centre fold and both wings and adjusted the folds so the two sides looked as even as possible.



My construction photo — Step 5

2. Why I chose this shape

My airplane has wide, roughly triangular wings and a folded centre body. I chose this shape because the wide wings may help it stay in the air at a slower speed. The centre folds make the paper stiffer. I also tried to keep both wings similar, because uneven wings can make the airplane turn or roll.

Measurements of my aircraft

Measurement	My aircraft
Paper size	A4 (29.7 cm × 21 cm)
Wingspan	11 cm
Aircraft length	16 cm
Wing area	44 cm²

Part B — Aerodynamics

1. How the airplane gets lift

When I throw the plane, air moves around its wings. The wings push some air downward and an upward force called lift is produced. Lift depends on speed, wing area, air density and the angle of the wings to the airflow. If I throw the plane too slowly, it may not get enough lift. If I point it too steeply upwards, it may stall and drop.

2. The four forces

Force	What it means for my paper airplane
Lift	The upward part of the aerodynamic force that helps the plane stay in the air.
Weight	The pull of gravity downwards.
Drag	Air resistance that slows the plane down.
Thrust	The push from my hand at launch. There is no continuous thrust because the plane has no motor.

At launch, my hand gives the plane speed. During the glide, gravity makes it move downwards while lift helps support it. Drag uses up the plane's motion energy, so the plane eventually comes down.

3. Wing area, wing loading, centre of gravity and stability

Wing area: My measured wing area is 44 cm². The wing area helps determine how much lift can be produced.
Wing loading: This means weight divided by wing area. My wing area is 44 cm², but I have not measured the aircraft's mass, so I cannot calculate its numerical wing loading. A lower wing loading can help a plane fly more slowly, but it does not guarantee longer flight by itself.
Centre of gravity: This is the balance point of the airplane. If the nose is too heavy, the plane may dive. If the balance point is too far back, it may pitch up and stall.
Stability: Both wings should be similar and the centre fold should be straight. This helps the plane fly without turning or rolling too much.

4. Why it may stay in the air longer than a dart

A normal paper dart often has narrow wings and flies faster. My airplane has wider wings, which may help it glide at a lower speed. A lower sink rate means losing height slowly, which is useful for longer airtime. Maximum glide distance is different: it means travelling the greatest horizontal distance for a given loss of height. My design must be tested to know whether it actually stays airborne longer.

Part C — Experiment

1. Flight-time results

I tested my paper airplane five times. The recorded flight times were 4, 4, 5, 5 and 5 seconds.

Mean flight time calculation

Mean flight time = (4 + 4 + 5 + 5 + 5) ÷ 5
Mean flight time = 23 ÷ 5
Mean flight time = 4.6 seconds

The shortest flight was 4 seconds and the longest flight was 5 seconds. The mean flight time was 4.6 seconds. The results show that the airplane stayed in the air for about four to five seconds in these trials.

2. Two changes I could test

Change 1 — Small wing-tip adjustment: I could make a small, equal fold at both wing tips to see if the plane becomes more stable. I would keep the launch conditions the same and compare five flights before and after the

change.

Change 2 — Adjust the nose folds: If the plane repeatedly pitches up or dives, I could adjust the existing nose folds slightly to change its balance. I would change only this feature and compare the average flight time. I would not add clips, tape or extra paper.

Conclusion

I made a broad-wing paper glider from one A4 sheet. Its wingspan is 11 cm, its length is 16 cm and its wing area is 44 cm². In my five trials, the flight times were 4, 4, 5, 5 and 5 seconds, giving a mean of 4.6 seconds. The wide wings may help the plane glide slowly, but the balance, wing alignment and launch technique also affect its performance. I can use the mean flight time to compare future changes to the design.

Sources and assumptions

The design description is based on my own construction photos and the project instructions. The explanations of lift, drag, weight, thrust and wing loading are based on basic aerodynamics concepts. The dimensions and flight times in this report are the measurements provided for my aircraft.